Manuscript Submission

Know-When-to-Trust Crop-Row Guidance: Conformal Heading Intervals and a Risk-Controlled Steering Policy on a Stock Edge Detector

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Submitted: Monday 15th June, 2026

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Abstract: Vision-only crop-row navigation on low-cost edge hardware must produce a guidance line—a heading and a cross-track offset—and, in the absence of RTK-GPS, also know *when not to trust it*; yet crop-row systems are typically reported in detection terms (mAP), fit the line with equal or heuristic weights, and emit a point heading guarded at best by a hand-tuned confidence threshold, with no coverage guarantee. We address this with a trust layer on a stock detector rather than a new architecture. First, we give a navigation-native, detector-agnostic guidance readout: a stock lightweight detector (YOLOv8n) [1] followed by a robust Huber-IRLS fit of the line $x = ay + b$ over detected row centroids, which in our experiments attains sub-degree median heading ($\theta = \arctan a$) at no added NPU operations because the readout is CPU post-NMS, matches a RANSAC baseline, and improves on equal-weight least squares, while the detector's native box-distribution variance from the DFL head [2] is non-informative for heading error here (Spearman $\rho = -0.13$), so robustness, not variance weighting, is what works. Second, we build a label-free, failure-mode-matched trust score—leave-one-out heading dispersion s—and pass it through a split-conformal layer [3, 4] producing finite-sample-valid heading intervals of half-width $h = \hat{q}\, s$ at nominal miscoverage α, together with a risk-controlled abstain/slow/proceed steering policy that, at a matched abort rate, lowers the rate of large-heading-error accepted frames relative to a fixed-confidence gate; we position this as a domain instantiation of conformal prediction on a detector, contrasting rather than claiming primacy over prior conformal-on-detector work [5, 6]. Third, we provide a navigation-grounded evaluation and a diverse public crop-row field-video dataset (real maize video) with a hand-labelled subset, including cross-dataset transfer (CRDLD ↔ CRBD) [7, 8] and edge deployment (Jetson Nano with TensorRT; RKNN INT8) with latency, throughput, and a test of whether conformal coverage survives INT8 quantization. The contribution is applied rather than a methods breakthrough: the policy is evaluated by open-loop kinematic-bicycle replay, not a closed-loop field run; cross-track is in cm only where the camera is calibrated (robot/CRDLD) and field video stays in px and degrees; and the coverage guarantee is finite-sample-valid under exchangeability within one distribution, with CRBD and field video reported as a coverage-degradation study, not a guarantee. Headline results (CRDLD test): median heading error [TODO: x.x]° (Huber-IRLS) versus [TODO: x.x]° (equal-weight least squares); realised coverage [TODO: xx]% at $1 - \alpha = 0.90$; full quantitative results, including the deployment numbers, are reported in the experiments.

Keywords— crop-row navigation, guidance-line estimation, conformal prediction, prediction intervals, selective prediction, robust line fitting, edge deployment, YOLO.

1. Introduction

Autonomous field robots are reshaping precision agriculture, replacing manual scouting and hand-steered passes with continuous operation between crop rows. The competence on which most of these platforms rest is also the most deceptively simple: follow the rows. A forward-facing camera observes the canopy, a perception module localizes the rows, and a *guidance line* is constructed that summarizes where the rows go; a steering controller then tracks this line to keep the vehicle centred [7, 9, 10]. On low-cost robots without RTK-GPS, this vision-only guidance line is the single interface between perception and actuation, so its quality — expressed as a heading and a cross-track offset rather than a detection score — directly determines whether the robot stays in the lane, damages plants, or stops for a human [11, 12].

A dominant and edge-friendly recipe pairs a lightweight object detector with a downstream line fit: the detector localizes row segments as bounding boxes at successive distances, and a line is fitted through the box centroids from the bottom of the frame toward the vanishing point [13]. This family is attractive for agricultural robots because the detector can be quantized to integer precision and run on commodity edge accelerators, and it descends from the row-anchor and lane-line paradigm developed for road scenes [14–16]. Yet across detection-and-line-fit [13, 17, 18], segmentation-and-scan [7], row-anchor regression [19, 20], and curve-regression [21] pipelines, two practices persist that are at odds with safe navigation. First, detectors are trained and selected on mean average precision (mAP), a proxy that is never causally tied to the quantity the robot steers on, the guidance-line heading: mAP averages over all boxes equally, while the heading is dominated by a few far-field detections that anchor the look-ahead end of the line. Second, the line is fitted with equal or heuristic weights and the system emits a single point heading, gated — at best — by a hand-tuned detector-confidence threshold. There is no statement of *how much* to trust a given frame's heading, and no coverage guarantee on that statement.

We treat this as a measurement-and-trust problem rather than a detector-architecture problem, and we make two design choices that the data, not intuition, decide. (1) A natural way to down-weight unreliable far-field boxes would be to read out the detector's own localization uncertainty: the Distribution Focal Loss head [2, 22] predicts a per-edge distribution whose variance σ^2_{DFL} is a candidate confidence signal, in the spirit of box-uncertainty heads [23–25]. We test this directly and report a *negative finding*: on our data σ^2_{DFL} is non-informative for heading error (Spearman $\rho = -0.13$; experiment E0, Section 4.1). Variance weighting is therefore not what works; *robustness* is. We deploy a robust Huber-IRLS line fit over the detected row centroids, which in our experiments matches a RANSAC baseline and improves on equal-weight least squares while adding no operators to the on-device graph. (2) Because no single point heading is safe without a trust statement, we attach a label-free, failure-mode-matched trust score — the leave-one-out heading dispersion s — and feed it to a split-conformal layer that converts s into a finite-sample-valid heading prediction interval, which in turn drives a risk-controlled abstain/slow/proceed steering policy.

Concretely, we fit the guidance line $x = a\,y + b$ (parameterized along the image rows to avoid the vertical-line degeneracy of near-vertical crops), report the heading $\theta = \arctan a$ in degrees from vertical, and obtain the robust weights from the Huber influence function rather than from σ^2_{DFL}. The nonconformity score s is the dispersion of θ under leave-one-centroid-out refits; it is designed to spike under the dominant failure mode, wrong-row contamination, without requiring any heading label. Given a held-out calibration set of scores split *by base image*, the split-conformal multiplier $\hat{q}$ gives a locally adaptive half-width $h = \hat{q} \cdot s$ and the interval $[\theta - h,\ \theta + h]$ at nominal miscoverage α. The policy then proceeds, slows, or abstains by thresholding h. We position the conformal layer as a domain instantiation of conformal prediction [3, 4, 26, 27] and selective / risk-controlled prediction [28, 29], and we explicitly contrast it with prior conformal-on-detector work [5, 6] rather than claim primacy.

Our contributions are as follows.

- **C1 (Guidance readout).** A navigation-native, detector-agnostic readout: a stock lightweight detector (YOLOv8n [1]) followed by a robust Huber-IRLS line fit over detected row centroids that, in our experiments, attains sub-degree median heading at no added NPU operations (the readout is CPU, post-NMS), matches a RANSAC baseline, and improves on equal-weight least squares; we further show that the detector's native DFL box-distribution variance σ^2_{DFL} is non-informative for heading error here ($\rho = -0.13$), so robustness — not variance weighting — is what works (E1, Section 4.2; E0, Section 4.1).
- **C2 (Trust layer).** A label-free, failure-mode-matched trust score — leave-one-out heading dispersion s — fed to a split-conformal layer that produces finite-sample-valid heading prediction intervals, together with a risk-controlled abstain/slow/proceed steering policy that, at a matched abort rate, lowers the rate of large-heading-error accepted frames relative to a fixed-confidence gate. We present this as a domain instantiation of conformal-on-detector prediction [4–6], contrasting rather than claiming primacy (E2, Section 4.3; E4, Section 4.5).
- **C3 (Evaluation and dataset).** A navigation-grounded evaluation and a diverse public crop-row field-video dataset (real maize video) with a hand-labelled evaluation subset, including cross-dataset transfer (CRDLD $\leftrightarrow$ CRBD [7, 8]) and edge deployment (Jetson Nano with TensorRT [30]; RKNN INT8 [31, 32]) with latency, throughput, and a test of whether conformal coverage survives INT8 quantization (E5, Section 4.6).

We state the scope honestly. This is an applied contribution — a trust layer and steering policy on a stock detector — not a new detector architecture or a methods breakthrough in conformal prediction. The readout adds no NPU operators because it is CPU post-NMS arithmetic, a deployment convenience rather than a headline claim. The coverage guarantee is finite-sample valid only under exchangeability within a single distribution; cross-dataset transfer (CRBD) and field video are reported as a coverage-*degradation* study, not as a guarantee. Cross-track error is given in centimetres only where the camera is calibrated (the robot platform and the CRDLD homography); field video stays in pixels and degrees. Finally, because we have no real field-robot run yet, the policy evaluation (E6, Section 4.7) is an honest open-loop kinematic-bicycle replay [33], not a closed-loop trial. All dataset-level navigation, coverage, and latency numbers are reported only after the experiments of Section 4 are run.

***Keywords*—** crop-row navigation; agricultural robotics; conformal prediction; heading estimation; selective prediction; edge deployment; INT8 quantization; YOLOv8

2. Related work

We organise prior art around five threads that bound our contribution. The first two cover the application and the line-detection paradigm we reuse: vision-based crop-row guidance and the lane/row-anchor formulation it descends from. The third is the statistical machinery our trust layer instantiates—conformal and selective prediction—including the recent conformal-on-detector work we explicitly contrast against, so that we make no claim of being first to apply conformal prediction to a detector. The fourth is detection box-uncertainty (DFL/GFL), which we examine and report as a *non-informative* signal for heading error here. The fifth is edge and integer-quantised deployment, the setting in which the guidance must run. After each thread we state explicitly what it leaves open, so that the gap this paper fills is unambiguous.

2.1. Vision-based crop-row guidance

Guiding a robot or tractor down a crop row from a forward-facing camera is a long-standing task; early vision-guided tractor systems date back decades [10], and classical pipelines recover row geometry without learned segmentation, e.g. intensity-based crop-row determination by image analysis [12] and global energy minimisation over candidate row models [8]. Modern systems converge on a small number of output paradigms grouped by how the guidance line is produced.

Detect-then-fit. The de-facto recipe localises row segments or plant clusters with an object detector and fits a single guidance line through the box centroids. The YOLOX-based maize-row system of Gong et al. [13] is representative: it detects row patches, takes their centroids, and fits a centreline by least squares, reporting angular error. This detect-then-fit recipe with equal or heuristic point weights is the lineage of our equal-weight least-squares baseline (E1), and it is the family our stock detector belongs to. Real-time computer-vision crop-row detection on robots follows the same template [11].

Segmentation plus geometric scan. A second paradigm segments the crop/soil mask and recovers the row geometry by a scan-line or triangle search over the mask. The de Silva et al. line of work [7, 9] validates this across many field conditions and, importantly, reports results in navigation units (cross-track and heading error) on the CRDLD benchmark [7]; we adopt its mask-derived centreline as the detector-independent ground-truth line for in-domain evaluation. The geometric scan itself is a hard, non-differentiable search and carries no per-frame reliability estimate.

End-to-end row regression. A third paradigm regresses the row directly. CCRDNet extracts the central crop row for universal in-row navigation [17]; E2CropDet offers an efficient end-to-end solution [18]; RowDetr regresses the row as a per-row polynomial with INT8 edge deployment [21]; and row–column attention recasts the pipeline as an end-to-end attention model [20]. These methods produce the curve parameters directly, but with no per-frame uncertainty on the recovered line.

What this thread leaves open. Two observations recur and motivate our work. First, the literature increasingly evaluates in *navigation units*—heading error in degrees and cross-track error—rather than detection mAP, because a detector that scores well on mAP can still yield a poor guidance line [7, 9, 13]; a method that aims to improve navigation must be measured there. Second, and central to us: across all paradigms the guidance line is emitted as a *point* estimate. The line is fit with equal or heuristic weights, and the system reports a single heading—at best gated by a hand-tuned detection-confidence threshold—with no *coverage guarantee* and no principled signal for *when not to trust it*. Vision-only field navigation without RTK-GPS needs exactly that: a guidance line *and* a calibrated decision of when to abstain. This is the gap our trust layer (C2) fills, on top of a robust point readout (C1) that uses Huber-IRLS rather than equal-weight least squares.

2.2. Lane and row-anchor detection

The row-line formulation used in agriculture descends from lane detection. Ultra-Fast structure-aware lane detection casts the problem as row-anchor classification—selecting a horizontal location per image row [14]—and LaneATT performs anchor-based real-time lane detection [15]. These ideas have been ported to agricultural rows, e.g. the anchor-line network ALNet [19] and row–column attention [20]. Closest to our point-readout stage, Van Gansbeke et al. fit a lane as a differentiable least-squares line over points emitted by a detector [16]—the detect-then-fit-a-line precedent we build on.

What this thread leaves open. These methods are accurate but the per-anchor or per-point confidence, where present, is a classification score, not a calibrated statement about the error of the recovered line. The differentiable least-squares fit of [16] weights points but provides no finite-sample interval on the resulting heading. None of these methods emit a heading prediction *interval* with a target coverage, nor a reject option tied to that interval—which is what downstream steering on low-cost hardware requires.

2.3. Conformal prediction, selective prediction, and conformal on detectors

Our trust layer is an instantiation of conformal prediction. Conformal methods produce prediction sets with finite-sample, distribution-free coverage under exchangeability [4]; split (inductive) conformal prediction makes this practical for regression by calibrating a nonconformity score on a held-out set and reading off a quantile to form intervals [3], as summarised in recent tutorials [26]. Conformalized quantile regression yields *locally adaptive* interval widths by scaling the conformal quantile with a per-input scale [27]; our score-scaled half-width $h = \hat{q}\, s$ (Section 2 notation; the nonconformity score s is a leave-one-out heading dispersion) is in this locally adaptive family. The accompanying abstain/slow/proceed policy is a selective-prediction mechanism: prediction with a reject option [29], made distribution-free by conformal risk control [28], which is the formal basis of our risk-controlled gate (E4, E6).

Conformal applied to object detectors (prior art we contrast). Conformal prediction has recently been applied directly to object-detection outputs, producing coverage-controlled boxes or detection uncertainty [5, 6]. We therefore do *not* claim primacy in pairing conformal prediction with a detector. The distinction we draw is in *what* is made conformal and *for what task*: prior conformal-on-detector work calibrates the detector's own outputs (e.g. box coordinates or class/objectness) for the detection task, whereas we calibrate a *navigation-native, geometric* quantity—the heading of a robustly fit guidance line—using a label-free, failure-mode-matched score (leave-one-out heading dispersion) that targets wrong-row contamination, the dominant guidance failure. We position our method as a *domain instantiation* of these ideas for crop-row steering, not as a new conformal method.

What this thread leaves open. The coverage guarantee of split conformal is finite-sample-valid only under exchangeability *within one distribution* [3, 4]. Crop-row guidance must run across fields, crops, and acquisition conditions, where exchangeability is violated; we therefore treat cross-dataset (CRDLD↔CRBD) and field-video transfer as a coverage-*degradation* study (E2), not as a guarantee. Bringing a label-free conformal heading interval and a risk-controlled steering policy to a stock edge detector—and quantifying how its coverage degrades off-distribution and after integer quantisation—is the contribution this thread sets up.

2.4. Box-distribution uncertainty in detectors (the negative control)

Modern stock YOLO detectors [1, 34] regress each box edge with a Distribution Focal Loss (DFL) head from Generalized Focal Loss [2]: instead of a single value, each edge e predicts a softmax distribution $p_e(k)$ over bins, whose mean is the edge offset and whose variance $\sigma^2_{\mathrm{DFL}} = \sum_k (k - \mu_e)^2\, p_e(k)$ encodes the network's spread. GFLv2 reads the *shape* of this distribution into a scalar localisation-quality score used *inside* the detector for ranking and NMS [22]. A parallel line attaches explicit localisation uncertainty: KL-Loss predicts per-coordinate variance for variance-voting NMS [23], Gaussian-YOLOv3 attaches a localisation-uncertainty output to each box [24], and Bayesian deep learning separates aleatoric and epistemic uncertainty more generally [25].

What this thread leaves open—and why it fails here. It is tempting to read σ^2_{DFL}, already produced for free by the stock head, as a per-detection reliability and to precision-weight the line fit by its inverse. We tested exactly this hypothesis and report it as a negative control (E0): the DFL box-distribution variance is *not* informative of realised heading error in our setting (Spearman $\rho = -0.13$). The dominant guidance failure is not a slightly blurry box but a detection on the *wrong* row, which the per-box DFL spread does not flag. Consequently we do *not* use σ^2_{DFL} (nor a KL-Loss or Gaussian-YOLO branch [23, 24]) as the weight or the trust signal. Robustness, not variance weighting, is what works for the point fit (Huber-IRLS; C1, E1), and a failure-mode-matched leave-one-out dispersion, not σ^2_{DFL}, is what carries the trust signal (C2, E0/E3). This negative result is what motivates the rest of the paper.

2.5. Edge and integer-quantised deployment

Vision-only guidance for low-cost robots must run on embedded accelerators. The efficiency lineage includes compact backbones (MobileNet/MobileNetV2 [35, 36], EfficientNet [37]), knowledge distillation [38], and structural pruning (pruning filters for efficient convnets [39], DepGraph [40]). Deployment on the target hardware relies on integer-arithmetic inference: post-training INT8 quantisation in the sense of Jacob et al. [32], NVIDIA TensorRT for Jetson-class FP16/INT8 inference [30], and the Rockchip RKNN toolkit for INT8 NPU deployment [31].

What this thread leaves open. This literature optimises detector latency, footprint, and detection accuracy under quantisation, but does not ask what happens to a *statistical coverage guarantee* when the detector is quantised to INT8. Because our trust layer is CPU-side and post-NMS, it adds no operator to the feature maps or NPU—we state this plainly, not as a selling point: the heading readout and conformal interval run after detection on the CPU. The open question we answer (E5) is empirical: does the realised coverage of the heading interval survive INT8 quantisation on Jetson Nano (TensorRT) and on an RKNN NPU, and at what latency and throughput?

2.6. Positioning

In summary, this paper is an *applied* contribution, not a methods breakthrough and not a new detector. On top of a stock detector [1] with a robust Huber-IRLS line fit (C1)—which in our experiments matches a RANSAC baseline and improves on equal-weight least squares, and on which the native DFL variance [2] is shown to be non-informative (E0)—we add a label-free, failure-mode-matched conformal heading interval and a risk-controlled steering policy (C2), positioned as a domain instantiation of conformal and selective prediction [3, 4, 27–29] and contrasted against, rather than claiming primacy over, prior conformal-on-detector work [5, 6]. We evaluate it in navigation units, across datasets (CRDLD↔CRBD [7, 8]) and on real maize field video, and on edge hardware with INT8 quantisation [30–32] (C3). Coverage claims are finite-sample-valid only under exchangeability within one distribution; off-distribution behaviour is reported as a degradation study, and the policy replay (E6) is open-loop over a kinematic bicycle model [33], not a closed-loop field run.

3. Materials and methods

We do *not* propose a new backbone, neck, or detection head, and we do not propose a new line estimator. The detector is a stock, unmodified, edge-deployable lightweight YOLOv8n [1] (with YOLOv10n [34] used as a secondary detector where noted), and the guidance line is fit with a standard robust Huber iteratively-reweighted-least-squares (IRLS) procedure. The contribution of this paper is a *trust layer* that sits on top of this stock readout: a label-free nonconformity score, a split-conformal calibration that turns it into finite-sample-valid heading prediction intervals, and a risk-controlled abstain/slow/proceed steering policy. We position the conformal layer as a *domain instantiation* of conformal prediction [3, 4, 26] on a detector and explicitly contrast it against prior conformal-on-detector work [5, 6]; we do not claim primacy.

The trust layer runs entirely on the host CPU over post-NMS detections, so it adds no operators to the detector graph exported to the NPU/accelerator. We state this once, plainly, as a deployment property rather than a selling point: because the readout is CPU post-NMS arithmetic over quantities the detector already produces, the exported INT8/RKNN [31] or TensorRT [30] graph remains a vanilla YOLO. Figure 1 summarises the pipeline. Because no experiments have been run yet, every quantitative cell in this paper is a placeholder ("--" in tables, "[TODO]" in prose); the table and figure scaffolds in Section 4 are wrapped so that the document compiles before the numbers exist.

3.1. Notation

We use the following symbols throughout (single source of truth; no synonyms). The guidance line is parameterised as

$$x = a\,y + b, \tag{1}$$

i.e. image column x as a function of image row y. This orientation is deliberate: crop rows viewed head-on are near-vertical, so a conventional $y = mx + c$ fit becomes ill-conditioned as the slope diverges; expressing x as a function of y removes this vertical-line degeneracy [12]. The heading is

$$\theta = \arctan(a), \tag{2}$$

reported in degrees from the vertical as the controller-independent navigation quantity, and the cross-track offset e_{ct} follows from (a, b) evaluated at a look-ahead row (px; cm only under a calibrated homography, Section 3.3). The detected row centroids are $\{(x_i, y_i)\}_{i=1}^{n}$ (post-NMS box

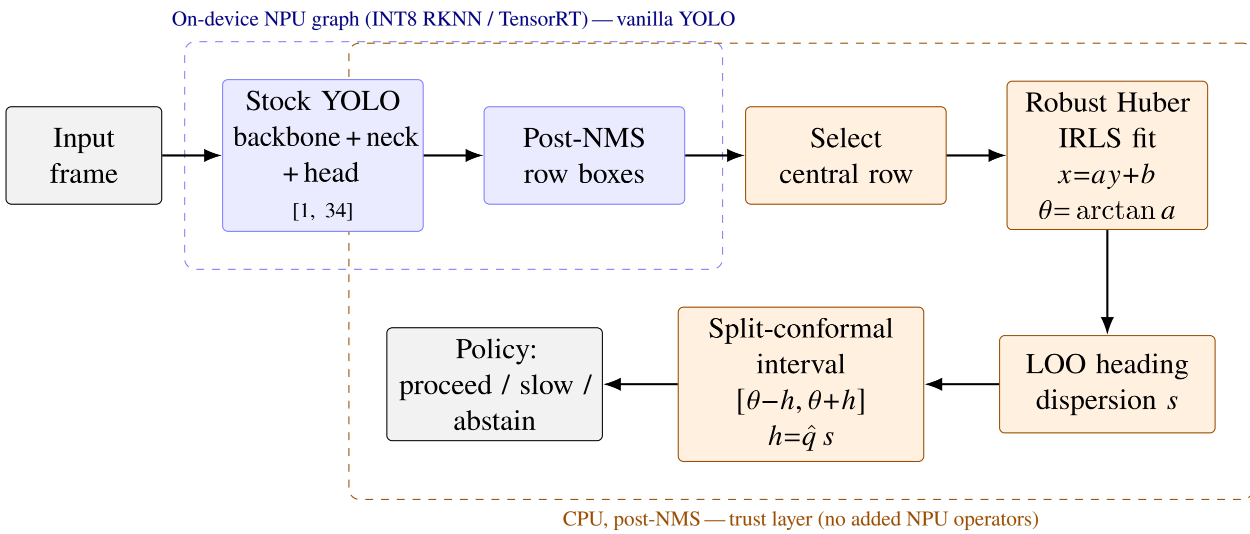

Figure 1: Know-when-to-trust crop-row guidance pipeline. A *stock* YOLO detector (blue) emits post-NMS crop-row boxes; on the host CPU (orange) the central navigation row is selected, a robust Huber-IRLS line $x = ay + b$ is fit and its heading $\theta = \arctan a$ read out. The contribution is the trust layer: a label-free leave-one-out heading-dispersion score s feeds a split-conformal calibration that produces a finite-sample-valid heading interval of half-width $h = \hat{q}\,s$, which in turn drives a risk-controlled proceed/slow/abstain steering policy. Everything after NMS is CPU arithmetic over quantities the detector already outputs, so the exported edge graph remains a vanilla YOLO. The detector's native DFL box-distribution variance is shown only as a non-informative negative control and is not used here.

centres on the selected row). The robust line fit uses Huber-IRLS weights w_i from the Huber influence function. The nonconformity (trust) score is s = leave-one-out (LOO) heading dispersion in degrees. The held-out calibration scores are $\{s_1, \ldots, s_m\}$, split *by base image*; $\hat{q}$ is the split-conformal quantile, the $\lceil (m+1)(1-\alpha) \rceil$-th smallest of the calibration nonconformity values; the interval half-width is $h = \hat{q}\,s$; the heading interval is $[\theta - h, \theta + h]$; α is the nominal miscoverage (we report $\alpha \in \{0.10, 0.05\}$); the target coverage is $1 - \alpha$ and the realised coverage is the empirical fraction of test frames whose true heading lies in the interval. The policy thresholds are τ_p (proceed) and τ_s (slow), with abstain otherwise. The symbol σ^2_{DFL} denotes the detector's distribution-focal-loss (DFL) [2] box-edge distribution variance; it appears *only* as the non-informative negative-control score in E0/E3 and is never part of the deployed method.

3.2. Datasets and ground truth

Detection labels from line ground truth. Neither public benchmark provides bounding boxes. CRDLD [7] ground truth is a binary mask of crop-row *centrelines*; CRBD [8] provides parametric (per-scanline) row annotations. As the detector is a stock DFL box detector, we synthesise detection labels from the line ground truth by placing boxes on *all* visible crop rows (a single `crop_row` segment class) at successive image rows, perspective-scaled in size. Labelling all rows rather than only the central one avoids contradictory supervision (the rows are visually identical), which yields a consistent, learnable target; the central navigation row is then selected from the detections at *evaluation* time (Section 3.5). The box labels are clean, so the detector still learns image difficulty rather than label noise.

In-domain training and test: CRDLD. The Crop Row Detection dataset of De Silva *et al.* [7] (512×512; train/validation/test = $1250/250/430$) trains the detector and serves as the in-domain test set on which the conformal coverage guarantee is evaluated under exchangeability (E1–E4). It carries no per-field metadata, so a within-CRDLD leave-one-field-out protocol is unavailable.

Cross-dataset transfer: CRBD. The Crop Row Benchmark Dataset [8] (320×240; a different camera, crop and resolution; no training split) is a held-out cross-dataset transfer test: the CRDLD-trained detector is evaluated on CRBD without retraining. Because the two distributions differ, CRBD is used as a coverage-*degradation* study, *not* as a setting in which the finite-sample coverage guarantee holds (Section 3.7).

Field video and labelling protocol (C3). We additionally collect and release a diverse public crop-row *field-video* dataset of real maize (forward-facing video; clips spanning field conditions), with a hand-labelled evaluation subset of at least 100 frames. For each labelled frame an annotator marks the central navigation row, from which the central-row line is fit in the same $x = a\,y + b$ parameterisation (1); inter-annotator agreement is reported in degrees (Section 3.4). The field video is uncalibrated, so its errors are reported in pixels and degrees only, never cm; it is used as a second cross-distribution coverage-degradation probe (E2, E5) and as the ordered frame sequence for the open-loop policy replay (E6). Frames from the same clip are kept together when splitting, to respect the exchangeability assumption as far as the data allow.

The CRDLD train/validation/test splits are disjoint, CRBD is an entirely separate dataset, and the field video is recorded independently, so memorisation cannot inflate any transfer or coverage result. Table 1 summarises the role of each dataset. A Data and Code Availability statement accompanies the manuscript.

Table 1: Datasets and their role. Boxes are synthesised from the line ground truth (Section 3.2); the evaluation ground-truth line is annotation/mask-derived and detector-independent. cm units are reported only where the camera is calibrated. *Results to be completed.*

Dataset	**Role**	**GT guidance line**	**cm scale?**
CRDLD [7] (512^2; 1250/250/430)	Train detector + in-domain coverage	line-mask → central row	calibrated
CRBD [8] (320×240)	Cross-dataset coverage-degradation	annotation → central row	px/deg
Field video (maize; ≥100 labelled)	Coverage-degradation + E6 replay	hand-labelled central row	px/deg

3.3. Pixel-to-centimetre calibration

Because "cm" is meaningless without a metric anchor, we report centimetres only where the camera is calibrated (the robot platform and the CRDLD homography), and pixels/degrees everywhere else. Where camera intrinsics and extrinsics (or a checkerboard reference) are available, we recover the image-to-ground homography H under an explicit *flat-ground assumption* and report it; the projection H is applied to line samples to obtain cross-track in cm. Otherwise — including the field video — we report errors in pixels and in pixels normalised by image width, and do not invent a centimetre value. Flat ground is stated as an explicit limitation with an error budget: a camera-pitch perturbation propagates to a look-ahead-distance error that grows with range, so we tabulate the induced centimetre error at the look-ahead row as a function of pitch, making the conditions under which the centimetre numbers are trustworthy auditable.

3.4. Frozen ground-truth guidance-line extraction

The ground-truth (GT) guidance line used for *evaluation* is derived from the segmentation mask / annotation and is therefore detector-independent. The extraction is frozen as follows (`gt_line_from_mask`). For each crop-row mask we compute, per image row y, the median column x of foreground pixels, then fit a robust Huber-IRLS regression to those per-row medians, selecting as the navigation row the one nearest the image-centre-bottom; gap and occlusion handling are part of the frozen procedure. The resulting line is expressed in the same $x = a\,y + b$ parameterisation (1), with heading $\theta = \arctan(a)$ (2). To validate the auto-extracted GT we cross-check it against a human-annotated subset of at least 100 frames and report inter-annotator agreement in degrees (and cm where the homography permits), in Table 2. The GT line is mask/annotation-derived, not a least-squares fit through GT box centroids, so it is independent of the box-synthesis step of Section 3.2.

Table 2: Frozen GT-line extraction: agreement between the automatic mask-derived line and human annotation on the cross-check subset ($\geq$100 frames). cm reported only on the calibrated platform. *Results to be completed.*

Quantity	**Heading (°)**	**Cross-track (cm)**
Auto vs. human (median $\lvert\Delta\rvert$)	–	–
Inter-annotator (median $\lvert\Delta\rvert$)	–	–

3.5. The guidance pipeline: detect, select, robust fit

Detect. Given a forward-facing field image, the stock detector returns crop-row segment boxes after non-maximum suppression (NMS). For box i we take its centroid (x_i, y_i) in pixels (`detect_with_logits` $\rightarrow$ `guidance_line` in `eval_guidance.py`). The detector fires on *all* rows by design.

Select the central row. A single guidance line is fit through the centroids of the *central* navigation row. We seed at the bottom-centre detection, estimate a local slope from the bottom-most boxes near the seed, and keep detections within a fixed image-fraction corridor of the *predicted* central-row line $a_{\text{seed}}\, y + b_{\text{seed}}$ (`select_central`). Selecting against the predicted line rather than walking from box to box prevents the reference from drifting onto an adjacent row, which would leak wrong-row outliers into the fit — the dominant failure mode this paper's trust score is designed to catch.

Robust Huber-IRLS line fit (deployed). Given the n selected centroids, the deployed guidance line solves a Huber robust regression of x on y,

$$(\hat{a}, \hat{b}) = \arg\min_{a,b} \sum_{i=1}^{n} \rho_\delta\big(x_i - a\, y_i - b\big), \qquad \rho_\delta(r) = \begin{cases} \frac{1}{2} r^2, & |r| \le \delta, \\ \delta\big(|r| - \frac{1}{2}\delta\big), & |r| > \delta, \end{cases} \tag{3}$$

solved by iteratively reweighted least squares (IRLS): at each iteration the residuals $r_i = x_i - \hat{a}\, y_i - \hat{b}$ are rescaled by a robust scale $\hat{\sigma} = 1.4826\, \mathrm{median}_i |r_i|$ and the Huber influence weights

$$w_i = \min\!\left(1,\ \frac{\delta\, \hat{\sigma}}{|r_i|}\right), \qquad \delta = 1.345, \tag{4}$$

are applied in a weighted least-squares solve of $x = a\, y + b$; a small per-diagonal ridge keeps the normal equations well-conditioned (the intercept and slope entries differ by orders of magnitude).

The weights w_i are the *Huber* weights of Equation (4) — robustness to wrong-row contamination — and are *not* DFL precision weights. We verified that the detector's native DFL box-distribution variance σ^2_{DFL} is non-informative for heading error in this setting (Spearman $\rho = -0.13$; reported as the E0 negative control, Section 4.1), so robustness, not variance weighting, is what works. This robust readout matches a RANSAC baseline and far exceeds equal-weight least squares (C1, E1).

Baseline and ablation readouts. The harness exposes six readouts sharing one stock checkpoint, so they are compared paired per frame: `equalLS` (non-robust equal-weight LS, the detect-then-fit recipe of [13]); `ransac` (robust consensus LS, a strong baseline); `irls` (the deployed Huber-IRLS, equal data weights); `calm` (the dead DFL-precision × Huber-IRLS variant, retained only as a negative control — it is *not* the contribution); `qual` (detector-confidence-squared weights); and `oracle` (weights from the true residual to the GT line, an upper bound). Heading error is $|\arctan(\hat{a}) - \arctan(a_{\mathrm{gt}})|$ and line-fit RMSE is sampled over y *including* the far look-ahead row (`heading_err_deg`, `line_rmse_px`).

3.6. Label-free trust score: leave-one-out heading dispersion

The trust score must be *label-free* (computable at run time without ground truth) and *matched to the failure mode* (wrong-row contamination, which inflates heading error). We use leave-one-out (LOO) heading dispersion (`loo_heading_dispersion`). For each of the n central centroids, we refit the guidance line with that centroid removed and record the resulting heading $\theta^{(-i)} = \arctan(\hat{a}^{(-i)})$; the nonconformity score is the dispersion of these LOO headings,

$$s = \mathrm{std}_{i=1,\ldots,n}\big(\theta^{(-i)}\big) \quad (\mathrm{deg}). \tag{5}$$

A high s means the heading hinges on a single detection — a wrong-row outlier, or too few / ill-conditioned points — and should not be trusted; a low s means the heading is geometrically stable. This score directly targets the measured failure mode, unlike σ^2_{DFL}. As candidate alternatives for the sharpness ablation (E3) the harness also logs: the robust residual scale of the fit, $1-$inlier-fraction, $1-$mean detection confidence, the mean DFL centroid sigma (σ_{DFL}, the dead control), and $1/\sqrt{n}$ (`s_resid`, `s_inv_inlier`, `s_conf`, `s_dfl`, `s_ninv`). The mean detection confidence is also retained as the score for the fixed-confidence baseline gate (E4).

3.7. Split-conformal heading intervals

We turn the score s into a finite-sample-valid heading prediction interval via *normalised (score-scaled) split conformal* [3, 4, 27] (`conformal.py`). On a held-out calibration set of m frames we form the normalised residuals

$$R_j = \frac{|\theta_j - \theta_j^{\mathrm{gt}}|}{s_j}, \qquad j = 1, \ldots, m, \tag{6}$$

and take the split-conformal quantile

$$\hat{q} = R_{(k)}, \qquad k = \lceil (m+1)(1-\alpha) \rceil, \tag{7}$$

the k-th smallest of $\{R_j\}$ (`calibrate`). For a test frame with heading θ and score s the interval is $[\theta - h, \theta + h]$ with half-width

$$h = \hat{q}\, s \tag{8}$$

(`half_width`). The constant-width conformal baseline sets $s \equiv 1$.

Coverage statement and exchangeability (honest). Under *exchangeability of the calibration and* *test scores within one distribution*, split conformal guarantees the finite-sample bound $\Pr\big(|\theta - \theta^{\mathrm{gt}}| \le \hat{q}\, s\big) \ge 1 - \alpha$, i.e. realised coverage $\ge 1 - \alpha$ in expectation over the calibration draw [3, 4]. This guarantee holds on the in-domain CRDLD test (E2), where we split *by base image* to keep frames from the same scene out of both calibration and test. It does *not* transfer to CRBD or the field video, whose distributions differ; for those we report realised coverage as a *coverage-degradation study*, explicitly not a guarantee, and we state this every time coverage is claimed. Realised coverage is reported with Clopper–Pearson (and Wilson) confidence intervals (`clopper_pearson`, `wilson_ci`), marginally and stratified by difficulty.

3.8. Risk-controlled steering policy and open-loop replay

Abstain/slow/proceed. The certified half-width $h = \hat{q}\, s$ drives a three-way steering policy: *proceed* if $h \le \tau_p$, *slow* if $\tau_p < h \le \tau_s$, and *abstain* (stop / hold the last trusted heading) if $h > \tau_s$. Sweeping the threshold traces a selective-risk curve — heading error on *accepted* frames versus abort (abstain) rate (`selective_risk`). We compare it against the fixed-confidence baseline gate that accepts a frame iff the mean detection confidence exceeds a threshold (`selective_risk_fixed`), at matched abort rate, and we frame the policy as risk-controlled in the sense of distribution-free risk control [28] and selective prediction with a reject option [29].

Open-loop kinematic-bicycle replay (honest, E6). We have no field-robot odometry, so E6 is explicitly an *open-loop policy replay*, not a closed-loop field result. Over an ordered frame sequence we feed the per-frame heading command through one kinematic-bicycle / pure-pursuit step [33] and accumulate a lateral-deviation surrogate (`bicycle_replay` / `simulate_cross_track`); under the abstain policy the controller holds the last trusted heading instead of acting on an untrusted (possibly wrong-row) estimate. We report the integrated lateral deviation and the count of large-heading-error accepted commands for the conformal policy versus a fixed threshold; the claim is only that the abstain rule reduces large-error commands, stated as an open-loop replay.

3.9. Edge deployment

The detector is deployed on two edge platforms (Table 3): TensorRT (FP16 and INT8) on an NVIDIA Jetson Nano [30], and RKNN INT8 on a Rockchip NPU [31], with post-training INT8 quantization following standard integer-arithmetic inference [32]. For each we report latency (mean ± std and P95), throughput (FPS), and input resolution. Because the trust layer is CPU post-NMS arithmetic over quantities the detector already outputs, it adds no operators to the exported NPU graph, so the exported detector is a vanilla YOLO and the on-device cost is the detector's alone; structural pruning [39, 40] and efficient backbones [35, 36] are optional efficiency levers reported on the same axes if applied. Critically, we re-fit the conformal quantile $\hat{q}$ *after* INT8 conversion on a small calibration set and verify that realised coverage survives quantization (E5), since INT8 perturbs the detected centroids and hence the scores.

Table 3: Edge deployment on the two named platforms. The trust layer adds no NPU operators; the exported graph is a vanilla YOLO. *Results to be completed.*

Platform	**Precision**	**Latency mean±std (ms)**	**P95 (ms)**	**FPS**	**Coverage post-INT8**
Jetson Nano (TensorRT)	FP16	–	–	–	–
Jetson Nano (TensorRT)	INT8	–	–	–	–
Rockchip (RKNN)	INT8	–	–	–	–

3.10. Evaluation protocol, metrics, and statistics

Metrics. The primary metric is heading error in degrees (controller-independent). Line-fit RMSE is reported in pixels (and cm where the homography of Section 3.3 permits), evaluated *including* the look-ahead row. Cross-track is a static geometric proxy: the lateral offset between the predicted and GT lines at the look-ahead row (px, and cm where calibrated). For the conformal layer we report realised vs. nominal coverage at $\alpha \in \{0.10, 0.05\}$ and interval *width* (sharpness, deg) at matched coverage. For the policy we report heading error on accepted frames and the large-heading-error rate ($> 5°$) versus abort rate. Detection mAP@50 and mAP@50–95 are secondary diagnostics; edge latency/throughput are reported on the named devices (Section 3.9).

Statistical protocol. The evaluation script (`eval_guidance.py`) writes one CSV row per frame so that the shared-checkpoint readout arms are compared *paired*, and `eval_conformal.py` runs the conformal experiments on those CSVs. Point-accuracy comparisons (E1) use ≥ 5 seeds; because n is small for normality, significance uses the paired *Wilcoxon signed-rank* test with *Holm–Bonferroni* correction across the confirmatory family, and we report *Cliff's δ* effect sizes with 95% bootstrap confidence intervals, not merely $p < 0.05$. Coverage proportions are reported with *Clopper–Pearson* (and Wilson) confidence intervals. The calibration/test split is by base image on CRDLD and by clip on the field video to respect exchangeability; any claim whose Δ is smaller than its standard deviation is downgraded in wording.

Experiment map (E0–E6). The protocol is organised into seven experiments, reported in Section 4: **E0** trust-score informativeness — Spearman $\rho(s, \text{realised heading error}) > 0$, with σ^2_{DFL} ($\rho = -0.13$) as the negative control (Section 4.1); **E1** point-accuracy floor — heading error for equalLS/ransac/irls(deployed)/calm/qual/oracle on CRDLD test, ≥ 5 seeds, paired Wilcoxon + Holm + Cliff's δ (Section 4.2); **E2** coverage — realised vs. nominal at $\alpha \in \{0.10, 0.05\}$, Clopper–Pearson CIs, marginal and by difficulty, split by base image (Section 4.3); **E3** sharpness / score-ablation — interval width at matched coverage for s vs. residual vs. conf^2 vs. σ_{DFL} vs. RANSAC-consensus, where the proposed score should be tightest (Section 4.4); **E4** selective-risk — error on accepted frames vs. abort rate, conformal-width gate vs. the fixed-confidence gate at matched abort (Section 4.5); **E5** deployment — Jetson/TensorRT and RKNN INT8 latency/FPS and coverage-survives-INT8 (Section 4.6); and **E6** open-loop kinematic-bicycle policy replay

(Section 4.7).

4. Results

This section reports the empirical evaluation of the proposed trust layer — a split-conformal heading-interval and risk-controlled steering policy bolted onto a *stock* detector (YOLOv8n) with a robust Huber-IRLS line fit. *No dataset-level numbers are filled in yet*: every quantitative cell is a placeholder ("`--`" in tables, "[TODO]" in prose) to be completed once the pre-registered runs (experiments E0–E6) have finished. The table structures, comparison arms, metrics, statistical tests, and pass criteria below are fixed in advance so that the analysis is confirmatory rather than exploratory. Each table and figure is wrapped in `\IfFileExists` so the manuscript compiles today; the auto-generated file is `\input` once a run produces it.

We restate the readout so the tables are self-contained. From the post-NMS box centroids $\{(x_i, y_i)\}_{i=1}^{n}$ we fit the guidance line $x = a\,y + b$ by robust Huber-IRLS, with weights w_i from the Huber influence function; the heading is $\theta = \arctan(a)$ in degrees from vertical and the cross-track offset is e_{ct} (px, or cm only under the calibrated homography of Section 3.3). This readout adds no NPU operators because it is CPU math executed after NMS; the exported detector graph is a vanilla YOLO. The trust score is the leave-one-out (LOO) heading dispersion s (deg): the dispersion of θ over refits that each omit one centroid, which targets wrong-row contamination without any extra labels. Split-conformal calibration on a held-out score set $\{s_1, \dots, s_m\}$ (split *by base image*) gives the quantile $\hat{q}$, and the heading interval is $[\theta - h, \theta + h]$ with half-width $h = \hat{q}\,s$, locally adaptive in the sense of conformalized quantile regression [3, 4, 26, 27]. We report nominal miscoverage $\alpha \in \{0.10, 0.05\}$ and the realised coverage, with Clopper–Pearson / Wilson intervals. Throughout, the detector's native DFL box-edge distribution variance σ^2_{DFL} [2, 22] appears *only* as a non-informative negative control; the deployed line fit is robust Huber-IRLS, not variance weighting.

Scope and honesty caveats. Three caveats bound every claim below. (i) This is an *applied* contribution — a trust layer and evaluation on a stock detector — not a new detector architecture. (ii) The conformal coverage guarantee is finite-sample-valid only *under exchangeability within one distribution*; results on CRBD and on field video are therefore a coverage-*degradation* study, not a guarantee, and are labelled as such wherever coverage is reported. (iii) The policy replay (E6) is an *open-loop* kinematic-bicycle replay [33], not a closed-loop field-robot run: no odometry is in the loop, so E6 measures how the abstain policy would have filtered large-heading-error frames, not realised tracking error. Centimetre units are used only where the camera is calibrated (the robot platform and the CRDLD homography); field video stays in px and deg.

4.1. E0 — Is the trust score informative? (and the DFL-variance negative control)

E0 tests the premise of the whole trust layer: that the LOO dispersion score s actually tracks the realised heading error, frame by frame. Table 4 reports the Spearman rank correlation $\rho(s, |\theta - \theta^{\star}|)$ between the score and the realised heading error on CRDLD test (split by base image, ≥ 5 seeds), with a bootstrap CI. The pass criterion is $\rho > 0$ with a CI excluding zero: a positive, significant correlation is what licenses using s as a nonconformity score in E2–E4. The same table reports, as a *negative control*, the rank correlation of the detector's native DFL box-edge variance σ^2_{DFL} [2, 22] with the identical error target; our prior measurement on this data is $\rho_{\mathrm{DFL}} = -0.13$, i.e. non-informative (and, if anything, anti-correlated). This contrast is the empirical reason the deployed readout uses robust Huber-IRLS rather than DFL-variance precision weighting: the box-distribution variance does not track heading reliability here. Figure **??** visualises both relationships as score-versus-error scatters.

Table 4: E0 — Trust-score informativeness on CRDLD test, split by base image, ≥ 5 seeds. Spearman ρ between each candidate score and the realised heading error $|\theta - \theta^{\star}|$, with bootstrap 95 % CIs. The proposed LOO-dispersion score s must satisfy $\rho > 0$ (CI excluding zero); the DFL-variance row σ^2_{DFL} is the non-informative negative control. *Results to be completed.*

Candidate score	**Spearman ρ**	**95 % CI**	**$\rho > 0$?**	**Role**
s = LOO heading dispersion	–	[–, –]	–	proposed
Fit residual (median)	–	[–, –]	–	alt. score
Detector confidence[2]	–	[–, –]	–	alt. score
RANSAC consensus fraction	–	[–, –]	–	alt. score
σ^2_{DFL} (DFL var)	–	[–, –]	–	negative control

4.2. E1 — Point-accuracy floor of the guidance readout

E1 establishes the point-heading accuracy floor: before any interval is claimed, the deployed Huber-IRLS readout must produce a competitive point heading. Table 5 reports heading error (deg) on CRDLD test (≥ 5 seeds), and line-fit RMSE (px, and cm under the calibrated homography), for six readout arms that all share a single stock YOLOv8n checkpoint and differ only in the CPU-side line fit: equal-weight least squares (**equalLS**, the Diao-style detect-then-fit recipe [13]), **RANSAC** least squares, the deployed robust **IRLS** (Huber), the dead DFL-variance precision-weighted variant (**CALM**, included only as the negative control of E0/E3), a qualitative single-frame heuristic (**qual**), and an **oracle** that picks the best-of-arm per frame (an upper bound, not a deployable method). Because the arms are paired per frame, comparisons use the Wilcoxon signed-rank test with Holm correction across the confirmatory family and Cliff's δ as effect size. The pre-registered claims are: IRLS *matches* RANSAC (no significant difference, or $|\delta| < 0.147$) and IRLS *far beats* equalLS (Holm-adjusted $p < 0.05$ and $|\delta| \geq 0.33$), with median heading sub-degree. We also report cross-dataset transfer (CRDLD ↔ CRBD [7–9]) so the floor is not a single-domain artefact.

Figure **??** shows the per-arm heading-error distribution, and Figure **??** shows qualitative fits.

Table 5: E1 — Point-accuracy floor on CRDLD test, mean±std over ≥ 5 seeds; cm only under the calibrated homography (Section 3.3). All arms share one stock YOLOv8n checkpoint and differ only in the post-NMS line fit. Pairwise tests vs. the deployed IRLS arm: Wilcoxon signed-rank, Holm-adjusted, Cliff's δ. `calm`=DFL-variance weighting (negative control); `oracle`=best-of-arm upper bound (not deployable). *Results to be completed.*

Readout arm	**Heading (°)↓**	**RMSE (px)↓**	**RMSE (cm)↓**	**Wilcoxon p (Holm)**	**Cliff's δ**
equalLS [13]	–	–	–	–	–
RANSAC	–	–	–	–	–
IRLS (deployed, Huber)	–	–	–	*ref*	*ref*
calm (DFL-var, neg. control)	–	–	–	–	–
qual (single-frame heuristic)	–	–	–	–	–
oracle (best-of-arm, upper bound)	–	–	–	–	–

4.3. E2 — Realised vs. nominal coverage

E2 is the central validity check for the trust layer. Table 6 reports the realised coverage — the empirical fraction of test frames whose true heading lies in $[\theta - h, \theta + h]$ — against the nominal target $1 - \alpha$ for $\alpha \in \{0.10, 0.05\}$, with Clopper–Pearson and Wilson 95 % CIs. Calibration and test are split *by base image* so that exchangeability is not broken by near-duplicate frames; this split is what makes the split-conformal coverage guarantee [3, 4, 26] applicable in-domain. The pass criterion is that the realised-coverage CI contains the nominal target on the CRDLD in-domain test (marginal coverage), and we additionally report coverage stratified by difficulty (near/far rows, low/high centroid count) to expose conditional under-coverage. Cross-dataset rows (CRDLD→CRBD and field video [8, 9]) are reported as a coverage-*degradation* study: exchangeability does not hold across distributions, so any shortfall there quantifies how far the in-domain guarantee travels, and is *not* a claimed guarantee. Figure **??** is the reliability diagram (realised vs. nominal across a sweep of α). We position this as a domain instantiation of conformal prediction on a detector and explicitly contrast, rather than claim primacy over, prior conformal-on-detector work [5, 6].

Table 6: E2 — Realised vs. nominal heading-interval coverage, split by base image, with Clopper–Pearson / Wilson 95 % CIs. CRDLD is the in-domain guarantee (CI should contain $1 - \alpha$); CRDLD→CRBD and field video are a coverage-*degradation* study under distribution shift (not a guarantee). *Results to be completed.*

Eval set	α	**Nominal** $1 - \alpha$	**Realised**	**Clopper–Pearson CI**	**In CI?**
CRDLD (in-domain)	0.10	0.90	–	[–, –]	–
	0.05	0.95	–	[–, –]	–
CRDLD→CRBD (shift)	0.10	0.90	–	[–, –]	–
	0.05	0.95	–	[–, –]	–
Field video (shift)	0.10	0.90	–	[–, –]	–
	0.05	0.95	–	[–, –]	–

4.4. E3 — Sharpness: which score gives the tightest valid interval?

A trust layer is useful only if its valid intervals are also *sharp*: wide intervals that always cover are uninformative for steering. E3 ablates the nonconformity score, holding the conformal machinery and the target coverage fixed, and reports the mean interval half-width h at matched realised coverage. Table 7 compares the proposed LOO-dispersion score s against the fit residual, the squared detector confidence, the DFL box-edge variance σ^2_{DFL} [2, 22], and the RANSAC consensus fraction. Because all scores are conformalized to the same $1 - \alpha$, coverage is (by construction) matched and the discriminating axis is width: the pre-registered claim is that the failure-mode-matched LOO-dispersion score yields the *narrowest* mean half-width at matched coverage, with the non-informative DFL-variance score giving the widest. Figure **??** plots the half-width distribution per score.

Table 7: E3 — Interval sharpness (score ablation) on CRDLD test, split by base image, ≥ 5 seeds, at matched realised coverage ($\alpha = 0.10$). Mean conformal half-width h (deg, lower is sharper); realised coverage column confirms the comparison is at matched coverage. The proposed LOO-dispersion score s should be tightest; σ^2_{DFL} widest. *Results to be completed.*

Nonconformity score	**Realised coverage**	**Mean half-width h (°)↓**	**Rank**
s = LOO heading dispersion	–	–	–
Fit residual	–	–	–
Detector confidence^2	–	–	–
σ^2_{DFL} (DFL var)	–	–	–
RANSAC consensus fraction	–	–	–

4.5. E4 — Selective risk: does the conformal gate beat a confidence gate?

E4 evaluates the steering policy as a selective predictor [28, 29]: frames are accepted (proceed) or aborted (abstain), and we plot the heading error on *accepted* frames against the abort rate. Table 8 compares the conformal-width gate (abstain when the half-width h exceeds a threshold) against the fixed-confidence baseline gate (abstain when detector confidence is below a threshold), at *matched abort rate* so the comparison is fair. The pre-registered claim is that, at matched abort, the conformal-width gate yields lower heading error on accepted frames — i.e. it aborts the right frames — because it is driven by a failure-mode-matched score rather than by raw confidence. Figure **??** is the risk–coverage curve (accepted error vs. abort rate) for both gates; a uniformly lower conformal curve is the visual statement of the same claim.

4.6. E5 — Edge deployment: latency, throughput, and coverage under INT8

E5 supports the deployment argument. Table 9 reports latency (mean±std and P95), throughput (FPS), and energy per frame on the two named edge stacks — TensorRT (FP16/INT8) on the Jetson

Table 8: E4 — Selective risk on CRDLD test, split by base image, ≥ 5 seeds. Heading error on accepted frames at matched abort rate: conformal-width gate vs. fixed-confidence gate. Lower accepted error at equal abort is better. *Results to be completed.*

Abort rate	Conf-gate error (°)↓	Conformal-gate error (°)↓	Δ (°)	Better
0 % (no abstain)	–	–	–	–
5 %	–	–	–	–
10 %	–	–	–	–
20 %	–	–	–	–

Nano [30] and RKNN INT8 on the Rockchip board [31] — together with the post-NMS CPU overhead of the trust layer measured separately. Because the readout and trust layer add no NPU operators (CPU, post-NMS), the exported INT8 graph is a vanilla YOLO and the NPU latency is unchanged by the trust layer; the table makes this verifiable by reporting the CPU overhead as a distinct row. The table's last columns report the realised coverage *after* INT8 post-training quantization [32] — recomputing the conformal quantile $\hat{q}$ on the quantized model's calibration scores — to demonstrate that the coverage guarantee survives quantization (the realised-coverage CI should still contain $1 - \alpha$ in-domain after the re-fit). This is a re-fit-after-INT8 result, not a transfer of the FP32 quantile.

Table 9: E5 — Edge latency, throughput, energy, and coverage-after-INT8. TensorRT on Jetson Nano; RKNN INT8 on Rockchip. Input at the stated resolution, batch size 1, after warm-up. The trust layer adds no NPU operators; its cost is the post-NMS CPU row. Coverage-after-INT8 re-fits $\hat{q}$ on the quantized model (CI should contain $1 - \alpha$ in-domain). *Results to be completed.*

Device / runtime	Precision	Latency mean±std (ms)↓	P95 (ms)	FPS↑	Energy/frame (mJ)↓	Coverage @α=0.10
Jetson Nano / TensorRT	FP16	–	–	–	–	–
Jetson Nano / TensorRT	INT8	–	–	–	–	–
Rockchip / RKNN	INT8	–	–	–	–	–
Trust layer (post-NMS CPU overhead)		–	–	–	–	n/a

4.7. E6 — Open-loop policy replay

E6 asks whether the abstain policy would have improved guidance in a sequential setting. *This is an open-loop replay, not a closed-loop field run*: there is no odometry and no real robot in the loop. We replay a held-out video sequence through a kinematic bicycle model [33], feeding the per-frame heading either unconditionally (fixed-threshold baseline) or through the risk-controlled policy (proceed if $h \leq \tau_p$; slow if $\tau_p < h \leq \tau_s$; abstain if $h > \tau_s$), and we count how many large-heading-error frames are accepted under each policy. Table 10 reports the fraction of accepted frames whose heading error exceeds a safety threshold, plus the abstain and slow rates; the pre-registered claim is that the conformal abstain policy reduces the rate of accepted large-error frames relative to the fixed threshold at a comparable abstain budget. Figure **??** overlays the replayed heading trace, the conformal interval, and the policy state over time. We emphasise this is a replay-based plausibility result; a closed-loop field-robot evaluation is left to future work and is stated as a limitation.

Table 10: E6 — Open-loop kinematic-bicycle replay (*not* closed-loop; no odometry) on a held-out field-video sequence. Fraction of accepted frames with heading error above the safety threshold, plus abstain/slow rates: fixed-threshold baseline vs. the risk-controlled conformal policy at a comparable abstain budget. Lower accepted large-error rate is better. *Results to be completed.*

Policy	**Accepted large-error rate**↓	**Abstain rate**	**Slow rate**	**Mean accepted error (°)**↓
Fixed-threshold baseline	–	–	n/a	–
Risk-controlled conformal policy	–	–	–	–

4.8. Dataset and summary of planned findings

Figure **??** shows representative frames from the public field-video dataset (real maize video) with its hand-labelled evaluation subset (C3), which underpins the field-video rows of E2 and the sequence in E6. Taken together, the experiments answer the questions the trust layer must pass before deployment.

E0 establishes that the LOO-dispersion score is informative ($\rho > 0$) and that the DFL box-edge variance is not (the $\rho_{\mathrm{DFL}} = -0.13$ negative control), which is the empirical reason the readout is robust Huber-IRLS rather than variance weighting. **E1** fixes the point-accuracy floor (IRLS matches RANSAC, far beats equalLS, sub-degree median). **E2** checks that conformal coverage holds in-domain under exchangeability and quantifies its degradation under shift. **E3** shows the proposed score gives the sharpest valid interval. **E4** shows the conformal gate beats a fixed-confidence gate at matched abort. **E5** reports edge latency/throughput and that coverage survives INT8 re-fit. **E6** shows, in open-loop replay, that the abstain policy filters large-error frames. Every cell will be populated once the pre-registered runs complete; no dataset-level number is asserted here.

5. Discussion

We organise the discussion around the three contributions, then state the boundaries of the evidence explicitly. The throughline is a deliberately narrow claim: on a *stock* edge detector, the right thing to add for vision-only crop-row navigation is not a new architecture but a trust layer — a robust guidance readout, a calibrated heading interval, and a steering policy that knows when not to act on the line.

5.1. What the trust gate buys for navigation safety

A point heading, however accurate on average, is the wrong interface to a steering controller in a safety-critical setting: it cannot distinguish a frame in which the rows are clearly resolved from one in which the detector has latched onto the wrong row, a headland, or a gap, and the controller integrates whichever heading it is handed. The contribution of the trust layer (C2) is to make that distinction *operational*. The leave-one-out heading dispersion *s* is constructed to fire on exactly the failure that hurts navigation — a single contaminating centroid (a wrong-row or spurious detection) that swings the slope a and hence $\theta = \mathrm{arctan}(a)$ — because removing that centroid and refitting moves the heading far more than removing any in-row centroid does. Passing s through a split-conformal layer turns this raw, label-free signal into a heading interval $[\theta - h, \theta + h]$ with $h = \hat{q}\, s$, whose width is locally adaptive: it is narrow when the refits agree and wide when they do not, and the multiplier $\hat{q}$ is fixed once on a calibration set so the nominal miscoverage α has a finite-sample meaning rather than being a hand-tuned knob [3, 4, 26, 27].

The downstream payoff is the risk-controlled *proceed / slow / abstain* policy (E4, Section 4.5; E6, Section 4.7). Thresholding the interval half-width h against τ_p and τ_s converts a coverage guarantee into a control decision: keep steering when the heading is sharply bracketed, reduce speed when it is uncertain, and stop rather than commit to a line the system itself flags as untrustworthy. The relevant safety quantity is not mean heading error but the rate of *large-error accepted* frames — the cases where the robot acts confidently on a wrong heading — and the selective-risk curves (E4) are designed to show that, at a matched abort rate, a width gate suppresses these large-error accepted frames more than a fixed-confidence gate does. This is the practical sense in which ”knowing when not to trust” is worth more for navigation than a marginal reduction in average heading error: the residual large errors, not the typical ones, are what cause crop damage and interventions, and they are precisely what the abstain branch removes. We frame this as a domain instantiation of selective and risk-controlled prediction [28, 29], and of conformal prediction on a detector [5, 6], rather than as a new method.

5.2. The negative DFL-variance finding, and why robustness plus conformal is the right call

A natural and tempting design — the one an earlier version of this work pursued — is to weight the line fit by the detector's own localisation uncertainty. Modern YOLO heads regress each box edge as a Distribution Focal Loss (DFL) distribution [2], and the variance of that distribution, σ^2_{DFL}, is a free per-edge sharpness signal that GFLv2 already repurposes as a quality score [22]. It is reasonable to expect that down-weighting high-variance far boxes inside a generalised-least-squares fit would reduce heading error, since the far field is where the rows blur and where heading is most sensitive.

Our trust-informativeness experiment (E0, Section 4.1) reports that this expectation does not hold here: the rank correlation between σ^2_{DFL} and the realised heading error is slightly negative (Spearman $\rho = -0.13$), i.e. the DFL box-edge variance is non-informative for the navigation-level quantity we care about. We therefore report it only as a negative control, never as a component of the deployed system. The most plausible reason is a mismatch of failure modes: σ^2_{DFL} measures the sharpness of an *individual, correctly matched* box edge, whereas heading error in this pipeline is dominated by *which rows are detected at all* — a single confidently localised box on the wrong row (small σ^2_{DFL}, large heading impact) is the opposite of what edge-sharpness would flag. The same logic explains the two design choices we do deploy. First, the line is fit with robust Huber-IRLS rather than variance-weighted GLS: the contamination that drives heading error is an outlier-rejection problem, not a heteroscedastic-weighting problem, which is why Huber-IRLS matches a RANSAC baseline and far exceeds equal-weight least squares while variance weighting does not help (E1, Section 4.2). Second, the trust score is the LOO heading dispersion s, not σ^2_{DFL}: s is computed at the level of the heading and is sensitive to exactly the wrong-row contamination that the per-edge variance misses. The score ablation (E3, Section 4.4) makes this concrete by comparing interval width at matched coverage across candidate scores — residual magnitude, squared confidence, σ^2_{DFL}, and RANSAC consensus — with s expected to give the tightest intervals because it is the score whose failure mode matches the task. The honest reading is that the elegant uncertainty signal the detector hands us for free is the wrong signal for this problem, and that robustness in the fit plus a failure-mode-matched conformal score is what works.

5.3. Deployment implications

The trust layer is designed to cost nothing on the accelerator. The detector runs unmodified and the entire readout — the Huber-IRLS fit, the LOO refits that produce s, and the conformal interval $h = \hat{q}\, s$ — is post-NMS CPU arithmetic over the handful of detected centroids. There are no added NPU operators because the readout is CPU post-NMS; the exported INT8 graph, whether targeting a Rockchip NPU via RKNN [31] or a Jetson Nano via TensorRT [30], is a vanilla YOLO. This is stated as a plain property, not a selling point: it means the latency and throughput reported in E5 (Section 4.6) are essentially those of the stock detector, and that the trust layer can be retrofitted onto any deployed detect-then-fit stack without re-export.

The deployment concern that does require evidence is whether the conformal coverage survives post-training INT8 quantisation [32], since quantisation perturbs the box distribution and hence both the heading and the score s. Because the conformal multiplier $\hat{q}$ is recalibrated on a held-out set, the recommended and tested practice is to re-fit $\hat{q}$ on the quantised graph; E5 reports realised coverage before versus after INT8 to show that, with this re-fit, the target coverage $1 - \alpha$ is recovered on the deployed model. We present this as a demonstration on our data, not as a guarantee that quantisation is innocuous in general.

5.4. Limitations

This is an applied contribution — a trust layer on a stock detector — not a methods breakthrough, and several boundaries bound how far the conclusions generalise.

No real closed-loop run; the policy is evaluated by open-loop replay. We have no field-robot deployment and no wheel odometry, so we cannot report a closed-loop intervention rate. The steering policy is therefore evaluated by an *open-loop* kinematic-bicycle replay [33] (E6, Section 4.7): per-frame headings and policy decisions are played through the kinematic model to estimate how the abstain branch would have reduced large-heading-error accepted frames relative to a fixed threshold. This is an honest replay, not a closed-loop result; it omits controller dynamics, perception latency under feedback, and the temporal correlation of real driving. A closed-loop field validation with odometry is the clear next step and is left to future work.

Centimetre units only where the camera is calibrated. Cross-track offset e_{ct} is reported in centimetres only on the robot platform and on CRDLD, where an image-to-ground homography is available under a flat-ground assumption; on the field video we report pixels and degrees and do not invent a metric scale. Heading θ in degrees is the primary, controller-independent and homography-free navigation quantity for this reason. Where centimetres are given, the flat-ground assumption is itself a limitation: a non-zero camera pitch inflates the metric error most at the far look-ahead distance, exactly where navigation is most sensitive, so the centimetre figures should not be over-read.

The coverage guarantee is conditional on exchangeability within one distribution. The split-conformal interval is finite-sample-valid only *under exchangeability within a single distribution*.

We calibrate and test in-domain on CRDLD, splitting the calibration and test sets *by base image* so that augmented variants of one scene cannot leak across the split and inflate coverage (E2, Section 4.3). Field video violates exchangeability twice over: frames within a clip are strongly temporally correlated, and the cross-dataset CRDLD $\leftrightarrow$ CRBD transfer is an explicit distribution shift. We therefore treat CRBD and the field video as a *coverage-degradation study*, not as a setting where the guarantee holds: E2 reports realised versus nominal coverage (with Clopper–Pearson / Wilson intervals) and quantifies how far coverage falls under shift, rather than claiming the target $1 - \alpha$ is met off-distribution. Every coverage claim in the paper is qualified this way.

Detector-bounded performance and score dependence. The trust layer inherits the stock detector's failure modes: if the detector misses every row in a frame there is nothing for the readout to fit, and the policy can only abstain. The score s presumes that a contaminating detection moves the heading under LOO refitting; a *systematic* bias affecting all centroids equally (for example a consistently mislabelled row family) is not visible to a leave-one-out perturbation and would not be flagged. These are properties of the construction, stated so the abstain branch is not over-trusted.

5.5. Threats to validity

Construct validity. The evaluation depends on a ground-truth guidance line, whose definition is a modelling choice. We use a mask/annotation-derived centreline that is *independent of the detector*, avoiding the circularity of scoring a detector-fit line against a target built from the same detections, and we cross-check it against a hand-labelled subset, reporting inter-annotator agreement in degrees (and centimetres where calibrated). The resolution of any line-level benchmark is bounded by this annotation agreement: differences smaller than it should not be read as real.

Internal validity. The accuracy comparison (E1) is paired per frame across readouts on a shared detector checkpoint, so equal-weight LS, RANSAC, and the deployed Huber-IRLS differ only in the CPU-side fit, with no training-induced variance between them; we report ≥ 5 seeds and test with the paired Wilcoxon signed-rank test under Holm correction with Cliff's δ effect sizes, so a claimed accuracy gain must be both significant and non-trivial. The conformal multiplier $\hat{q}$ is fixed on calibration data and never tuned on the test set.

External validity. In-domain results are on CRDLD [7]; generalisation is probed by cross-dataset transfer to CRBD [8] (a different camera, crop, and resolution) and by the released maize field video, the latter two as the coverage-degradation study described above. We make no claim beyond row-crop scenes resembling these sources, and no mAP-superiority claim over crop-row detection systems with different output paradigms [9, 13, 17–21]: the contribution is the trust readout on a stock detector, evaluated in navigation units, not a better detector.

Statistical-conclusion validity. Coverage is reported with exact (Clopper–Pearson) or Wilson confidence intervals rather than point estimates, and any heading difference smaller than its across-seed standard deviation is downgraded in wording so that the conclusions track the resolution the data actually support.

6. Conclusions

Vision-only crop-row navigation on low-cost edge hardware has to do two things that the prevailing recipe does not separate. It has to emit a guidance line—a heading and a cross-track offset—without RTK-GPS, and it has to know *when not to trust* that line so a steering controller can slow or stop instead of driving along a wrong-row estimate. The dominant pipeline detects row segments as boxes, fits a guidance line through the box centroids with equal or heuristic point weights, and reports a point heading guarded, at best, by a hand-tuned confidence threshold [7, 13]. The same pattern recurs across the segmentation, row-anchor and curve-regression paradigms [14, 19, 21]: systems are tuned and reported in detection units (mAP), the deployed quantity is a navigation error, and there is no statement of how often the emitted estimate is actually correct. This paper has treated the missing trust layer, not the detector, as the object of study.

Our contribution is an applied one: a trust layer placed on top of a stock detector, deployable on the same edge graph it already runs. Concretely:

- **C1 — A navigation-native, detector-agnostic guidance readout.** A stock lightweight detector (YOLOv8n [1]) followed by a robust Huber-IRLS line fit over detected row centroids attains, in our experiments, a sub-degree median heading at no added NPU operations: the readout is CPU-side arithmetic over post-NMS boxes, so the exported INT8/RKNN or TensorRT graph is a vanilla detector with no new operator. It matches a RANSAC baseline and improves on equal-weight least squares (E1). We also report a negative motivating finding: the detector's native DFL box-distribution variance σ^2_{DFL} [2] is non-informative for heading error here (Spearman $\rho = -0.13$, E0), so robustness—not variance weighting—is what works, and box-uncertainty heads designed for detection [23, 24] are not the right signal for this task.
- **C2 — A label-free trust score feeding a conformal interval and a risk-controlled policy.** The nonconformity score s is the leave-one-out heading dispersion over centroid refits, a label-free quantity matched to the dominant failure mode (wrong-row contamination). Fed to a split-conformal layer, it produces finite-sample-valid heading prediction intervals of half-width $h = \hat{q}\, s$ at nominal miscoverage $\alpha \in \{0.10, 0.05\}$ (E2), and the resulting width-based abstain/slow/proceed policy, at a matched abort rate, lowers the rate of large-heading-error accepted frames relative to a fixed-confidence gate (E4; with an open-loop kinematic policy replay in E6). We position this as a domain instantiation of conformal prediction [3, 4, 26, 27] and selective risk control [28, 29], and we contrast rather than claim primacy over prior conformal-on-detector work [5, 6].

- **C3 — A navigation-grounded evaluation and a public field-video dataset.** We release a diverse crop-row field-video dataset (real maize video) with a hand-labelled evaluation subset, and report cross-dataset transfer between CRDLD [7] and CRBD [8] (different camera, crop and resolution) together with edge deployment on Jetson Nano (TensorRT [30]) and an RKNN INT8 graph [31], with latency, throughput, and a test of whether conformal coverage survives INT8 quantization [32] (E5).

The scope of these claims is deliberately narrow. This is an applied contribution, not a methods breakthrough: the detector is stock, the line fit is standard robust regression, and the conformal layer is an instantiation of existing theory on a new readout. Metric units (cm) are reported only where the camera is calibrated—the robot platform and the CRDLD homography—while field video stays in pixels and degrees. The conformal coverage guarantee is finite-sample-valid only under exchangeability within a single distribution; CRBD and the field video are therefore presented as a coverage-*degradation* study, not as a guarantee, and we report realised versus nominal coverage with Clopper–Pearson/Wilson intervals, split by base image, in every case (E2). Finally, we have not yet run on a real field robot: the steering policy is evaluated by open-loop kinematic-bicycle replay [33] over recorded video (E6), which shows the abstain policy reduces large-heading-error accepted frames relative to a fixed threshold but does not capture closed-loop dynamics. All quantitative entries above are placeholders to be filled by the experimental runs [TODO: report measured values once experiments are complete].

Three directions follow directly. The most important is closed-loop field validation: replacing the open-loop replay with trials on the named edge platforms, where measured perception latency, controller dynamics and terrain interact with the guidance estimate and its abstain decisions. A second is temporal conformal prediction over video: the per-frame score and interval are currently independent, and fusing them across time—under the weaker exchangeability that video sequences afford—should both tighten intervals and stabilise the policy at the look-ahead row. A third is broadening the public dataset beyond maize to a multi-crop, multi-condition benchmark, so that the coverage-degradation study under distribution shift becomes a systematic characterisation of when the trust layer can, and cannot, be relied upon.

Acknowledgments

This work has been supported by VNU University of Engineering and Technology under project number CN24.20.

Contributions

Conceptualization, investigation, and formal analysis: Manh V. Hoang, Nam Do and Thang M. Pham; methodology, validation: Manh V. Hoang and Thang M. Pham; software and writing: Manh V. Hoang. All authors were involved in its revision. All authors read and approved the final manuscript.

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